

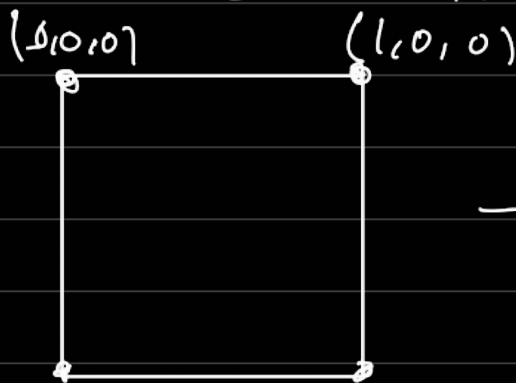
Hot big bang:

- Lines in spectra of nearby gal. have longer λ than lab.
- Un. is expanding, acc
- CMB: Blackbody $\sim$ uniform
- Expanding for only a certain amount of time B.B.

Hubble: $V = H_0 D$

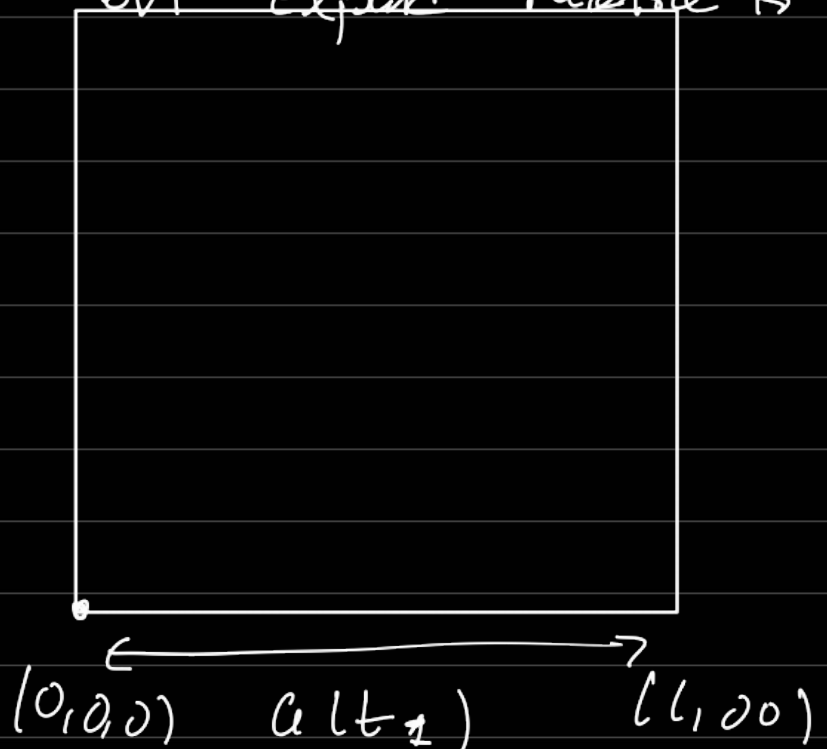
$$|H = \dot{R}/R$$

* Scale factor: All points in the universe are the "same" points but expand relative to one another



t_0

→



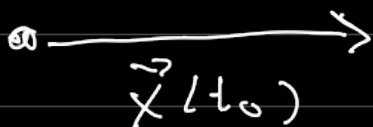
$(0,0,0)$

$a(t_1)$

$(1,0,0)$

$t_1 > t_0$

t_1



$$\vec{X}(t) = R(t) \vec{X}(t_0)$$

$t > t_0$

t_0

• $R(t)$: cosmic scale factor

$$\vec{X}(t) = R(t) \vec{X}(t_0) = \frac{\vec{R}(t)}{R(t_0)} \vec{X}(t_0)$$

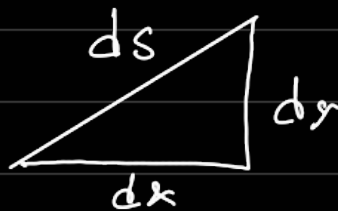
why? $V = HD$

$\uparrow$ vel. $\nwarrow$ Distance

$$H(t) = \frac{\dot{R}(t)}{R(t)}$$

* Describing spaces:

2D $\mathbb{R}^2$:



$$ds^2 = dx^2 + dy^2$$

3D: $ds^2 = dx^2 + dy^2 + dz^2$

2D: sphere:

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

$$dx = dr \cos(\theta) - r \sin(\theta) d\theta$$

$$dy = dr \sin(\theta) + r \cos(\theta) d\theta$$

$$dx^2 + dy^2 = dr^2 + r^2 d\theta^2$$

* Relativity: we measure things relative to the speed of light.

→ A photon will travel a distance

$$D^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 = (c \Delta t)^2$$

so

$$(c \Delta t)^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 = 0 = \underline{ds^2}$$

⇒ This is the "invariant distance"

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2$$

time - space

$$d\vec{x}^2 \equiv dx^2 + dy^2 + dz^2$$

$$+ : d\vec{x}^2 + dt^2 \Rightarrow x^2 + t^2 = 1 \quad S^3$$

$$- : dx^2 - dt^2 \quad x^2 - t^2 = 1$$

$$0 : d\vec{x}^2$$

* Homogeneous + Isotropic $\Rightarrow$ Curvature
can only be $(0, +, -)$

Cosmological Principle:

at any particular time the universe looks the same from all positions in space at a particular time and in all directions in space at any point are equivalent.

• We assume an obs has zero particular vel.

$d\Sigma^2$: space

$$\text{so since } \vec{x}(t) = R(t)\vec{x}(t_0)$$

Similarly

$$d\Sigma(t)^2 = R(t)^2 d\Sigma^2$$

$$ds^2 = c^2 dt^2 - R(t)^2 (dr^2 + S_k^2(t) (d\Omega^2))$$

$$S_k(r) = \begin{cases} r & k=0 \\ \sin(r) & k=1 \\ \sinh(r) & k=-1 \end{cases}$$

* Distances:

light travel distance

d_L

(i) proper distance d_p
 (ii) luminosity distance d_L
 (iii) Angular distance d_A
 $z \gg 1 \quad d_p \neq d_L \neq d_A$

Comoving distance $R_{or} \quad R_0 = R(t_0)$

$$d_p(t) = R(t) r$$

Light:

$$ds^2 = 0 = (c dt)^2 - R(t) (dr^2 - S_k(r)^2 d\Omega^2)$$

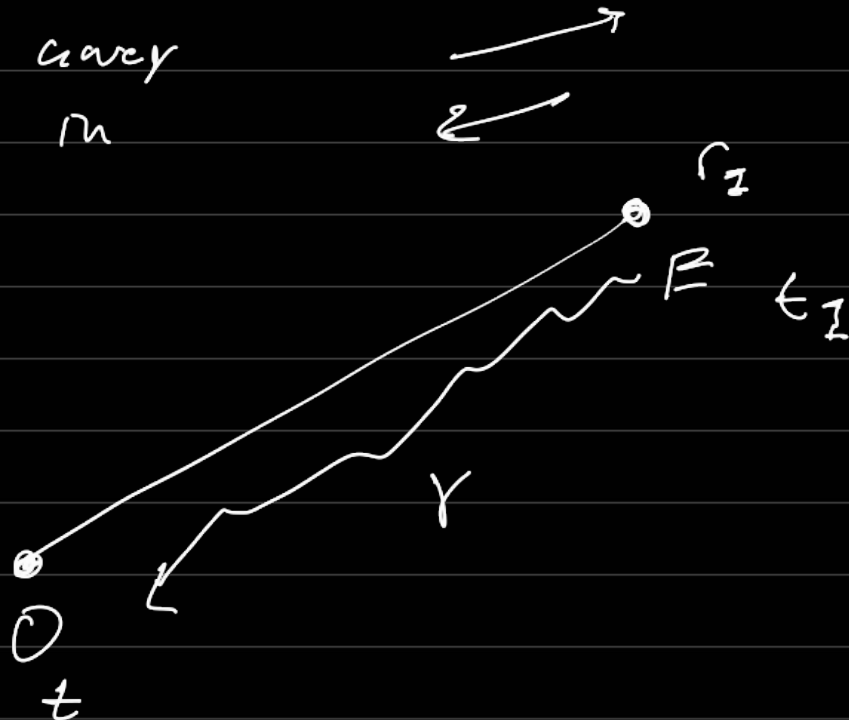
Assume γ moves radially so

$$d\theta = d\phi = 0$$

$$ds^2 = 0 = (c dt)^2 - R^2 dr^2$$

$$dr = \pm \frac{c dt}{R(t)}$$

$+$: moving away
 $-$: moving in



$$r_1 = c \int_{t_1}^t \frac{dt'}{R(t')}$$

$$t_1 \rightarrow 0$$

$$r_{ph} = c \int_0^t \frac{dt'}{R(t')}$$

• Marks the Boundary of the region about which $\odot$ can have learned anything by time t .

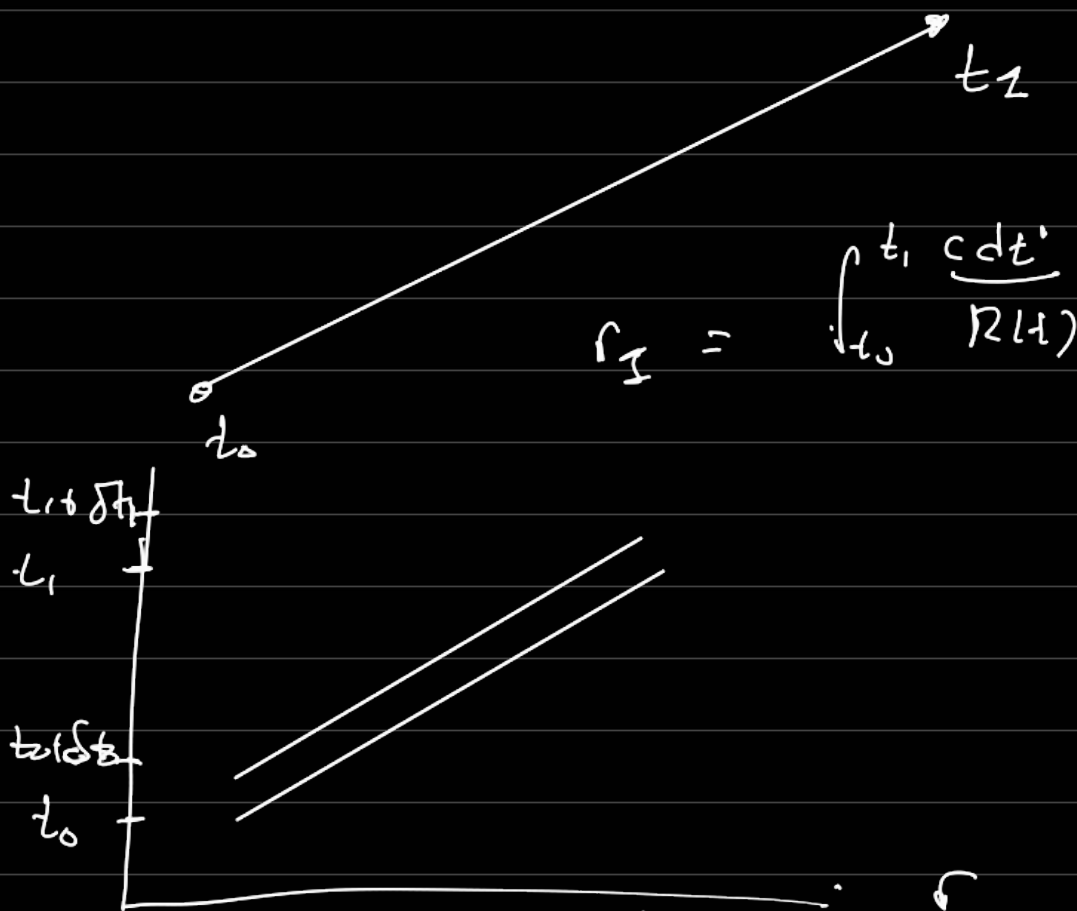
-> Particle Horizon.

$$R(t) = R_0 \left(\frac{t}{t_0}\right)^\alpha$$

$$r_{PH}(t) = c \int_0^t \frac{dt'}{R(t')^\alpha} = \frac{t_0^\alpha}{c R_0} \frac{1}{1-\alpha} t^{1-\alpha} \Big|_0^t$$

$$r_{PH}(t) = \frac{t_0^\alpha}{c R_0} \frac{t^{1-\alpha}}{1-\alpha} \quad \alpha < 1$$

Cosm. Redshift: $z \equiv \frac{\lambda_{obs}}{\lambda_{emitted}}$



$$r_I = \int_{t_0 + \delta t_0}^{t_1 + \delta t_1} \frac{c dt'}{R(t')}$$

$$\int_{t_0}^{t_1} \frac{c dt'}{R(t')} = \int_{t_0 + \delta t_0}^{t_0} \frac{c dt'}{R(t')} + \int_{t_0}^{t_1} \frac{c dt'}{R(t')} + \int_{t_1}^{t_1 + \delta t_1} \frac{c dt'}{R(t')}$$

$$\int_{t_0}^{t_0} \frac{c dt'}{R(t')} = 0$$

$$\int_{t_0+\delta t_0}^{t_0} \frac{cdt'}{R(t)} \approx \frac{c t_0}{R(t_0)} - \frac{c (t_0 + \delta t_0)}{R(t_0)}$$

$$-\frac{c \delta t_0}{R(t_0)} + \frac{c \delta t_1}{R(t_1)} = 0$$

$$\boxed{\frac{\delta t_0}{R(t_0)} = \frac{\delta t_1}{R(t_1)}}$$

• period : $f = \frac{1}{T}$
 $\lambda = c$
 $\lambda = \delta t_0 c$

$$\frac{\lambda_0}{R(t_0)} = \frac{\lambda_1}{R(t_1)} \Rightarrow \frac{\lambda_1}{\lambda_0} = \frac{R(t_1)}{R(t_0)}$$

$$z = \frac{\lambda_1 - \lambda_0}{\lambda_0} = \frac{\lambda_1}{\lambda_0} - 1$$

• $z = \frac{\lambda_1}{\lambda_0} - 1$ or $\boxed{z + 1 = \frac{\lambda_1}{\lambda_0}}$ *

$$z = \frac{\Delta \lambda}{\lambda_0}$$

Measures the fraction at which R has increased

$R(t_1)$: time today

$$R(t_1) \equiv R_0$$

$$R(t_0) = R(t)$$

$$1 + z = \frac{R_0}{R(t)}$$

$$S-R: 1 + z = \sqrt{\frac{1 + v/c}{1 - v/c}}$$

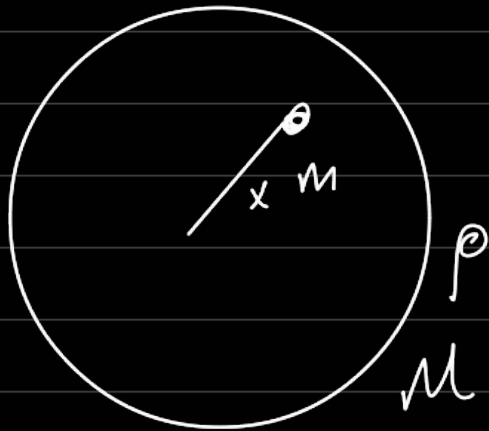
$$dp(t) = R(t) \dot{r} = \frac{\dot{R}}{R} dp = H(t) dp(t)$$

$$t_0: V(t_0) = H_0 dp(t_0)$$

$$1 + z \approx 1 + \frac{\delta v}{c} \quad / \quad z \approx \frac{\delta v}{c}$$

$$\delta V \approx c^2 \quad (\text{as we saw})$$

Dynamics / Evolution: Friedmann Eqs



$$M = \frac{4\pi}{3} x^3 \rho$$

$$\begin{aligned} E_{\text{tot}} &= K - U = \frac{1}{2} m \dot{x}^2 - \frac{G M m}{x} \\ &= \frac{1}{2} m \dot{x}^2 - G \left(\frac{4\pi}{3} x^3 \rho \right) \frac{m}{x} \\ &= \frac{1}{2} m \dot{x}^2 - G \frac{4\pi}{3} x^2 \rho m \end{aligned}$$

$$\frac{2 E_{\text{tot}}}{m} = \dot{x}^2 - \frac{G 8\pi}{3} x^2 \rho = \text{const.}$$

$$x(t) = d_p(t) = R(t) r$$

$$\dot{R}^2 - \frac{G 8\pi}{3} R^2 \rho = \frac{\text{const.}}{r^2} = \text{const.}$$

$$\dot{R}^2 - \frac{G 8\pi}{3} R^2 \rho = \text{const.}$$

$$\nabla GR \text{ gives us const.} \quad -k c^2$$

$$\dot{R}^2 - \frac{G 8\pi}{3} R^2 \rho = -k c^2$$

$$\left(\frac{\dot{R}}{R} \right)^2 = \frac{G 8\pi}{3} \rho - \frac{k c^2}{R^2}$$

$$H^2 = \frac{8\pi G}{3} \rho - \frac{Kc^2}{R^2}$$

$K=0$: Critical density

$$\rho_c = \frac{3H^2}{8\pi G}$$

dimensionless density parameter

$$\Omega \equiv \rho / \rho_c$$

$$\Omega(t) = 1 + \frac{Kc^2}{R^2(t)}$$

Friedmann models:

$$H^2 = \left(\frac{\dot{R}}{R}\right)^2 = \frac{8\pi G}{3} \rho - \frac{Kc^2}{R^2}$$

• Λ CDM:

$$\rho_B, \rho_{DM}, \rho_{DE}$$

Perfect fluids: $P = w\rho c^2$

Matter: $w=0$, $mc^2 \gg K.E.$
 $P \sim v^2$, so low pressure
 $P \ll \rho c^2$ (Dust)

Radiation: $w = 1/3$

$$P = \frac{1}{3} \mathcal{E} = \frac{1}{3} \frac{E}{Vol}$$

Vacuum: $w = -1$

Matter: $\rho = \frac{m}{Vol}$

$$\rho \sim \rho_0 \left(\frac{R}{R_0} \right)^{-3}$$

$$(c) \quad \kappa = 0$$

$$\left(\frac{\dot{R}}{R} \right)^2 = \frac{8\pi G}{3} \rho = \frac{8\pi G}{3} \rho_0 \frac{R_0^3}{R^3}$$

$$\dot{R}^2 = \alpha \frac{1}{R} \Rightarrow \dot{R} = \sqrt{\alpha} \frac{1}{\sqrt{R}}$$

$$\frac{2}{3} R^{3/2} = \sqrt{\alpha} t$$

$$R = \alpha^{1/3} \left(\frac{2}{3} \right)^{2/3} t^{2/3}$$

$$R(t) = \left(\frac{8\pi G}{3} \rho_0 R_0^3 \right)^{1/3} \left(\frac{2}{3} \right)^{2/3} t^{2/3}$$

Einstein-de Sitter

$$R(t) \equiv R_0 \left(\frac{t}{t_0} \right)^{2/3}$$

$$\dot{R}(t) = \frac{2}{3} R_0 \frac{t^{-1/3}}{t_0^{2/3}}$$

$$H(t) \equiv \frac{\dot{R}}{R} = \frac{\frac{2}{3} R_0 \frac{t_0^{2/3}}{t^{1/3} t_0^{2/3}}}{R_0 t^{2/3}} = \frac{2}{3} \frac{1}{t}$$

$$H(t) = \frac{2}{3t}$$

$$H_0 = \frac{2}{3t_0} \quad | \quad t_0 = \frac{2}{3} \frac{1}{H_0}$$

$$EdS \simeq t \simeq 9.6 \text{ Gyr}$$

$\kappa \neq 0$ / Matter Dominated:

$$\bullet \text{ Conformal time: } d\eta \equiv \frac{cdt}{R(t)} \Rightarrow \frac{d\eta}{dt} = \frac{c}{R}$$

$$\dot{R} = \frac{dR}{dt} = \frac{d\eta}{dt} \frac{dR}{d\eta} = \frac{c}{R} \frac{dR}{d\eta}$$

$$\frac{\dot{R}^2}{R^2} = \frac{c^2}{R^2} \left(\frac{dR}{d\tau} \right)^2 \Rightarrow \frac{c^2}{R^2} \left(\frac{dR}{d\tau} \right)^2 = \frac{8\pi G}{3} \rho_0 \frac{R_0^3}{R^3} - \frac{Kc^2}{R^2}$$

$$c^2 R'^2 = \frac{8\pi G}{3} \rho_0 R_0^3 R - Kc^2 R^2$$

$$R_* \equiv \frac{4\pi G \rho_0 R_0^3}{3c^2}$$

$$\frac{R'^2}{(R_*)^2} = 2 \frac{R_* R}{R_*^2} - K \frac{R^2}{R_*^2}$$

$$\left(\frac{R'}{R_*} \right)^2 = 2 \left(\frac{R}{R_*} \right) - K \left(\frac{R}{R_*} \right)^2$$

$$d\tau = \frac{cdt}{R(t)} \Rightarrow \int c dt = \int R(\tau) d\tau$$

$$(c dt) = \int R(\tau) d\tau$$

• Plug in

Q2 PS7: Ω_0 $H_0^2 = \frac{8\pi G}{3} \rho_c$

$$\Omega_0 \equiv \rho / \rho_{c,0} \Rightarrow \rho = \Omega_0 \frac{3 H_0^2}{8\pi G}$$

$$\left(\frac{\dot{R}}{R} \right)^2 = \left(\frac{8\pi G}{3} \right) \left(\Omega_0 \frac{3}{8\pi G} H_0^2 \right) - Kc^2 / R^2$$

$$H^2 = \Omega H_0^2 - Kc^2 / R^2$$

$$H^2 (1 - \Omega) = - \frac{Kc^2}{R^2}$$

$$R(t) = \left(\frac{c}{H} \right) \left(\frac{K}{1 - \Omega} \right)^{1/2}$$

$$R(t_0) = R_0 = \frac{c}{H_0} \left(\frac{k}{\Omega_0 - 1} \right)^{1/2}$$

$$R_* = 4\pi 6 R_0^3 \rho_0 / 3c^2 ; \text{ plug in}$$

Ex B3 $H^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{R^2}$

① EdS: $k=0$, matter dom. $\rho_m = \rho_0 \left(\frac{R_0}{R} \right)^3$

$$\frac{R_0}{R(t)} = 1 + z$$

$$\rho_m = \rho_0 (1+z)^3$$

$$H^2 = \frac{8\pi G}{3} \rho_0 (1+z)^3 \quad \Bigg| \quad \frac{8\pi G}{3} \rho_0 \equiv H_0^2$$

$$H(t) = H_0 (1+z)^{3/2}$$

② $t_{ph}(z) = \int_0^z \frac{cdt'}{R(t')}$

$$\begin{aligned} R_0 t_{ph}(z) &= \int_0^z \left(\frac{R_0}{R(t')} \right) c dt' \\ &= c \int_0^z \frac{R_0}{R(t')} dz' \end{aligned}$$

* $\frac{R_0}{R(t)} = 1 + z$

$$-1 \frac{R_0}{R(t)^2} \dot{R} dt = dz$$

$$- \left(\frac{R_0}{R(t)} \right) \left(\frac{\dot{R}}{R(t)} \right) dt = dz$$

$$= - \left(\frac{R_0}{R(t)} \right) dt = \frac{dz}{H}$$

$$t' = 0: \quad z \rightarrow \infty$$

$$t' = t: \quad z \rightarrow z$$

$$= -C \int_{\infty}^z \frac{dz}{H(z)} = C \int_z^{\infty} \frac{dz}{H(z)} \quad \checkmark$$

$$* R_{\text{hor}}(z) = C \int_{z_{\text{rec}}}^{\infty} \frac{dz}{H_0 (1+z)^{3/2}}$$

$$R_{\text{hor}}(z) = \frac{C}{H_0} (-2) (1+z)^{-1/2} \Big|_{z_{\text{rec}}}^{\infty}$$

$$R_{\text{hor}}(z) = \frac{2C}{H_0} (1+z_{\text{rec}})^{-1/2}$$

$$* R_{\text{hor}}(z) \Theta \approx D$$

$$D = d_A(z) = \frac{2c}{H_0 (1+z)}$$

$$\Theta = \frac{c}{H_0 (1+z)^{1/2}} = \frac{2c}{H_0 (1+z)}$$

$$\Theta = (1+z)^{-1/2}$$

$$\textcircled{1} \quad c_s^2 = dp/d\rho$$

$$H(z_{\text{rec}}) = H_0 (1+z_{\text{rec}})^{2/2}$$

$$t \approx \frac{1}{H(z_{\text{rec}})} \quad \text{Look up..}$$

$$* R_{\text{hor}}(z_{\text{cmb}}) = \frac{C}{H_0} \int_{1+z}^{\infty} dy \, y^{-1/2}$$

$$* R(t):$$

$$R(t) = R(t_{\text{rec}}) \left(\frac{a(t)}{a(t_{\text{rec}})} \right) = R(t_{\text{rec}}) (1+z)^{-1}$$

$$k=1: k(R) = kR_0(1 - \cos(R))$$

→ Means at some point $R \rightarrow 0$

$$\left\{ \begin{array}{l} \Omega > 1 \quad \rho > \rho_c \\ \Omega = 1 \quad \rho = \rho_c \\ \Omega < 1 \quad \rho < \rho_c \end{array} \right\} \rightarrow \text{Big Crunch}$$

$k=0$: or $k=-1$ R increases forever

* Radiation-dominated Universe:

• photons #:

$$n_\gamma \sim R^{-3}$$

$$E_\gamma = pc = h\nu = \frac{hc}{\lambda}$$

$$E_\gamma \sim R^{-1}$$

$$\rho c^2 = n_\gamma E_\gamma \sim R^{-4}$$

(K.E. > Rest)

$$\rho_R = \rho_{0,R} \left(\frac{R_0}{R} \right)^4$$

$$\left| \frac{\lambda_0}{R_0} = \frac{\lambda}{R} \right.$$

$$\lambda \sim \frac{R \lambda_0}{R_0}$$

$$\left(\frac{\dot{R}}{R} \right)^2 = \frac{8\pi G}{3} \rho_0 \left(\frac{R_0}{R} \right)^4 - \frac{kc^2}{R^2}$$

* Early times:

$$R(t) \sim t^{2/3} \quad \text{at early times}$$

* Want to know the transition from
Radiation $\rightarrow$ Matter

$$\rho_m(z) = \rho_{0,m} (1+z)^3$$

$$\rho_r(z) = \rho_{r,0} (1+z)^4$$

$$z = z_{eq}: \rho_m(z_{eq}) = \rho_r(z_{eq})$$

$$\rho_{0,m} = \rho_{r,0} (1+z_{eq})$$

$$\text{or } \Omega_{0,m} = \Omega_{0,r} (1 + z_{eq})$$

$$1 + z_{eq} = \frac{\rho_{m,0}}{\rho_{r,0}} = \frac{\Omega_{m,0}}{\Omega_{r,0}}$$

$$* \text{CMB: } T_{cmb} = 2.73 \text{ K}$$

$$z_{eq} = 10^9$$

$\Rightarrow R$ is small, so all times before z_{eq} dominate

$$\frac{1}{R^4} \text{ dominates over } \frac{1}{R^2}$$

$$\left(\frac{\dot{R}}{R}\right)^2 = \frac{8\pi G}{3} \rho_{r,0} \left(\frac{R_0}{R}\right)^4$$

$$\dot{R} = \sqrt{\alpha} R^{-1}$$

$$\frac{1}{2} R^2 \sim \sqrt{\alpha} t$$

$$R \sim t^{1/2}$$

$$E_{\infty} = -13.6 \text{ eV} = (13.6) (1.6 \times 10^{-19} \text{ J})$$

$$\approx 20 \times 10^{-19} \text{ J} \approx 2 \times 10^{-18}$$

$$\bullet E_{\gamma} \approx kT = 1.38 \times 10^{-23} \text{ J K}^{-1} T_{\gamma}$$

$$T_e = (1+z) T_0, \quad z_{\text{CMB}} \approx 1100$$

$$= (1100) 2.73 \approx 3000.3$$

$$\approx 3 \times 10^3$$

$$E_{\gamma} \approx 4.3 \times 10^{-20}$$

$$\underline{\text{Q3:}} \quad R(n) = \kappa R_* (1 - C_k(n)) / c t(n) = \kappa R_* (n - S_k(n))$$

$$\Omega_0 = 1.1, H_0 = 70 \text{ km}^{-1} \text{ Mpc}^{-1}$$

$$\begin{cases} \Omega > 1: & k = 1 \\ \Omega = 1: & k = 0 \\ \Omega < 1: & k = -1 \end{cases}$$

$$R(\eta) = R_* (1 - \cos(\eta)) = R_* (1 - \cos(\eta))$$

$$\frac{R_0}{R_*} = 1 - \cos(\eta_0) \Rightarrow 1 - \frac{R_0}{R_*} = \cos(\eta_0)$$

$$\eta_0 = \cos^{-1}\left(1 - \frac{R_0}{R_*}\right)$$

$$\eta_0 \approx 0.613$$

$$t(\eta_0) = \frac{R_*}{c} (\eta_0 - \sin(\eta_0)) =$$

$$t_0 = 0.654 H_0^{-1}$$

$$* \frac{dR}{dt} = \frac{dR}{d\eta} = 0 \quad | \quad R(\eta) = R_* (1 - \cos(\eta))$$

$$\eta = \pi$$

$$R(\eta = \pi) = 2R_* \Rightarrow R_0 = \frac{2}{11} R_*$$

• 750 Gyr

Acceleration Eq:

$$\bullet \text{ First law of thermo: } \frac{dS}{dt} = 0$$

→ No heat loss $\Rightarrow$ No temp gradient

$$T dS = dE + p dV$$

$$\bullet \frac{T dS}{dt} = \frac{dE}{dt} + p \frac{dV}{dt} = 0$$

* Matter:

$$E = M c^2 = c^2 \rho \frac{4\pi}{3} R^3$$

$$\bullet \frac{dE}{dt} = \frac{4\pi}{3} c^2 (\dot{\rho} R^3 + 3 \rho R^2 \dot{R})$$

$$V = \frac{4\pi}{3} R^3$$

$$\frac{dV}{dt} = 4\pi R^2 \dot{R}$$

$$0 = \frac{4\pi c^2}{3} (\dot{\rho} R^3 + 3 \rho R^2 \dot{R}) + 4\pi p R^2 \dot{R}$$

$$0 = (c^2 \rho + P) R^2 \ddot{R} + \frac{c^2}{3} \dot{R}^3$$

$$0 = 3\left(\rho + \frac{P}{c^2}\right)\left(\frac{\dot{R}}{R}\right) + \dot{\rho}$$

Fluid Equation:

$$\dot{\rho} + 3\left(\frac{\dot{R}}{R}\right)\left(\rho + \frac{P}{c^2}\right) = 0$$

$$\left(\frac{\dot{R}}{R}\right)^2 = \frac{8\pi G}{3} \rho - \frac{Kc^2}{R^2}$$

$$\dot{R}^2 = \frac{8\pi G}{3} \rho R^2 - Kc^2$$

$$2\dot{R}\ddot{R} = \frac{8\pi G}{3} (\dot{\rho} R^2 + 2\rho R \dot{R})$$

$$\dot{R}\ddot{R} = \frac{4\pi G}{3} \left[-3\left(\frac{\dot{R}}{R}\right)(\rho + \frac{P}{c^2}) R^2 + 2\rho R \dot{R} \right]$$

$$\ddot{R} = \frac{4\pi G}{3} \left(-3R\left(\rho + \frac{P}{c^2}\right) + 2\rho R \right)$$

$$\ddot{R} = \frac{4\pi G}{3} R \left[-\rho - \frac{3P}{c^2} \right]$$

$$\ddot{R} = -\frac{4\pi G}{3} R \left(\rho + \frac{3P}{c^2} \right)$$

$$\bullet GR \quad \nabla^2 \phi = 4\pi G \left(\rho + \frac{3P}{c^2} \right)$$

- Normal matter: $P \ll \rho c^2$

Vacuum:

* Einstein's argument in Newtonian gravity:

$$\nabla^2 \phi = 4\pi G \rho$$

• Constant density ρ should imply constant ϕ

$$\nabla \phi = -\vec{a}$$

• So zero acceleration means

$$\nabla \phi = 0$$

• So must correct $\nabla^2 \phi = 4\pi G \rho$

$$\nabla^2 \phi + \lambda \phi = 4\pi G \rho$$

• Static solution: $\phi = \frac{4\pi G \rho}{\lambda}$

* Modern: $\nabla^2 \phi = 4\pi G (\rho - \rho_{\text{rep}})$

$$\rho_{\text{rep}} = \lambda \phi / 4\pi G$$

$$\lambda \longleftrightarrow \rho_{\text{rep}}$$

• C.C. is equivalent to const. en. density with a repulsive grav effect

• Present even for $\rho_m = 0$, $\rho_{\text{rep}} = \text{Vacuum en.}$

$\Omega_{V,0} \equiv \text{Vacuum energy}$

$$\Omega_{V,0} = \rho_{V,0} / \rho_{\text{crit}} \sim \left(\frac{\Lambda}{\Lambda_0} \right)^4$$

* 1st law $Tds = dE + PdV$

$$E = mc^2 = \rho_v \frac{4\pi}{3} R^3 c^2$$

$$0 = d(\rho_v R^3 c^2) + p_v dR^3$$

• $d\rho_v = 0$

$$dR^3 (\rho_v c^2 + p_v) = 0$$

equation of state

$$p_v = -\rho_v c^2$$

$$\ddot{R} = -\frac{4\pi G R}{3} \left(\rho + \frac{3(-\rho_v c^2)}{c^2} \right)$$

$$\ddot{R} = \frac{8\pi G R}{3} \rho_v \quad (\text{Vacuum})$$

$\rho_v \text{ const.} \Rightarrow \ddot{R} = \alpha R$

$$R(t) = A \cosh(at) + B \sinh(at)$$

* $\ddot{R} > 0$ Univ. dominated by Vacuum is acc.

acceleration: $(\rho + 3p/c^2) < 0$

$$\Rightarrow w < -2/3$$

Dark Energy: Refers to any fluid with $w < -2/3$

$$\rho \propto R^0$$

$$\ddot{R}^2 = \frac{8\pi G}{3} \rho_v R^2 - Kc^2$$

K=0:

$$R(t) \sim e^{\pm Ht}, \quad H = \sqrt{\frac{8\pi G \rho_{vac}}{3}}$$

$$R^2 = K^2 R^2 - Kc^2 \quad (K=1, -1)$$

$$R(t) \sim \cosh(Ht) \quad (K=1)$$

$$R(t) \sim \sinh(Ht) \quad (K=-1)$$

→ All cases: Vacuum dominated will have expansion at late times

Multi-component cosmo models:

$$\rho(R) = \rho_m + \rho_r + \rho_v = \rho_{m,0} (1+z)^3 + \rho_{r,0} (1+z)^4 + \rho_v$$

$$H^2 = \frac{8\pi G}{3} (\rho_{m,0} (1+z)^3 + \rho_{r,0} (1+z)^4 + \rho_v) - \frac{Kc^2}{R^2}$$

$$\rho_{m,0} = \Omega_{m,0} \rho_{c,0}$$

$$\rho_{r,0} = \Omega_{r,0} \rho_{c,0}$$

$$\rho_v = \Omega_v \rho_{c,0}$$

$$H^2 = \frac{8\pi G}{3} (\Omega_{m,0} (1+z)^3 + \Omega_{r,0} (1+z)^4 + \Omega_v) \rho_{c,0} - \frac{Kc^2}{R^2}$$

$$3H_0^2$$

$$\rho_c = \frac{3H_0^2}{8\pi G}$$

$$\Rightarrow H^2 = H_0^2 \left(\Omega_{m,0} \left(\frac{R_0}{R} \right)^3 + \Omega_{r,0} \left(\frac{R_0}{R} \right)^4 + \Omega_v \right) - \frac{Kc^2}{R^2}$$

$$H_0^2 = H_0^2 \left(\Omega_{m,0} + \Omega_{r,0} + \Omega_v \right) - \frac{Kc^2}{R_0^2}$$

$$1 = (\Omega) - \frac{H_0^2 R_0^2}{c^2}$$

$$(1 - \Omega) = - \frac{K c^2}{H_0^2 R_0}$$

$$R_0^2 H_0^2 (1 - \Omega) = -K c^2$$

$$H^2 = H_0^2 \left(\right) + \frac{R_0^2 H_0^2 (1 - \Omega)}{R^2}$$

$$H^2 = H_0^2 \left[\Omega_{m,0} \left(\frac{R_0}{R} \right)^3 + \Omega_{r,0} \left(\frac{R_0}{R} \right)^4 + \Omega_{v,0} + (1 - \Omega) \left(\frac{R_0}{R} \right)^2 \right]$$

$$H^2 = H_0^2 \left[\Omega_{m,0} \left(\frac{R_0}{R} \right)^3 + \Omega_{r,0} \left(\frac{R_0}{R} \right)^4 + \Omega_{v,0} + (1 - \Omega_{m,0} - \Omega_{r,0} - \Omega_{v,0}) \left(\frac{R_0}{R} \right)^2 \right]$$

• Note: powers in $R^0, R^{-2}, R^{-3}, R^{-4}$
but none in R^{-1}

→ Could H be a formal Laurent series?

$$H(R) = \sum_{n=-\infty}^0 \frac{a_n}{R^n}$$

→ Asymptotically:

(R^0) : Vacuum energy $\rightarrow \langle \Omega | \Omega \rangle$

(R^{-2}) : Global topology

Q: Do black holes change the Global top?

(R^{-3}) : matter

(R^{-4}) : Radiation

Q: what if we tried adding R^{-1}

ex:

$$H^2 = \frac{8\pi G}{3} \rho_{a,0} \left(\frac{R_0}{R} \right) - \frac{K c^2}{R^2}$$

$$\dot{R}^2 = \frac{8\pi G}{3} \rho_{a,0} R_0 R - K c^2$$

$$\int \frac{dR}{\sqrt{aR - K c^2}} = \int dt$$

$$\left(\frac{2}{a}\right) \sqrt{\alpha R - kc^2} = t$$

$$\Rightarrow \alpha R - kc^2 = (\alpha t/2)^2$$

$$\alpha R = kc^2 + \alpha^2/4 t^2$$

$$R(t) = \alpha^{-1} (kc^2 + \alpha^2/4 t^2)$$

$$R_1(t) = \left(\frac{kc^2}{\alpha} + \alpha t^2 \right)$$

$$\dot{R}_1 = 2\alpha t$$

$$H = \frac{\dot{R}_1}{R} = \frac{2\alpha t}{\left(\frac{kc^2}{\alpha} + \alpha t^2 \right)}$$

$$k=0: H = \frac{2}{t} \quad \left| \quad t_0 = \frac{2}{H_0} = 9 \text{ Gyr} \right.$$

$$= 9 \times 10^9 \text{ yrs}$$

$$H_0 = 2/\tilde{t}_0 \Rightarrow \tilde{t}_0 = 27 \times 10^9 \text{ yrs}$$

→ twice as old

Q: R^{-5} :

$$H^2 = \frac{8\pi G}{3} \rho_{R,0} \left(\frac{R_0}{R} \right)^5 - \frac{kc^2}{R^2}$$

$$\dot{R}^2 = \frac{8\pi G}{3} \rho_{R,0} \frac{R_0^5}{R^3} - kc^2$$

$$\dot{R} = \sqrt{\frac{8\pi G}{3} \rho_{R,0} \frac{R_0^5}{R^3} - kc^2}$$

$$u = 1/R$$

$$du = -1/R^2 dR = -u^2 dR$$

$$\frac{dR}{dt} = -\frac{1}{u^2} \frac{du}{dt}$$

$$-\frac{1}{u^2} \frac{du}{dt} = \sqrt{\frac{8\pi G}{3} \rho_{R,0} R_0^5 u^3 - kc^2}$$

$$\int \frac{du}{u^2 (8u^3 - \gamma)^{1/2}} = - \int dt$$

$$k=0:$$

$$\dot{R} = \beta R^{-3/2}$$

$$R^{3/2} dR = \beta dt$$

$$\frac{2}{5} R^{5/2} = \beta t$$

$$R = \omega t^{2/5}$$

$$\dot{R} = \frac{2}{5} \omega t^{-3/5}$$

$$\rightarrow (\dot{R}/R) = \frac{\frac{2}{5} \omega t^{-3/5}}{\omega t^{2/5}} = \frac{2}{5} \left(\frac{1}{t} \right)$$

$$\text{flat: } \sum_n \frac{1}{n} = 2 \sum_n n^{-1}$$

$$= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

$$\approx \int dn n^{-1} = \int_1^N \frac{1}{n} dn \approx \ln(N)$$

$$H \sim 2 \ln(N) \frac{1}{t}$$

→ Must be cut and not summed over as $\ln(N)$ - terms already over estimates.

* For the current universe: $\Omega_{r,0} \approx 0$
from CMB

$$H^2 = H_0^2 \left(\Omega_{m,0} \left(\frac{R_0}{R} \right)^3 + \Omega_{v,0} + (1 - \Omega_{m,0} - \Omega_{v,0}) \left(\frac{R_0}{R} \right)^2 \right)$$

* $(\Omega_{m,0}, \Omega_{r,0}, \Omega_{v,0}, H_0)$

• Probes:

- (a) Type Ia supernovae
- (b) CMB
- (c) Gal. clusters
- (d) light elements

$$\text{Redshift time: } 1+z = \frac{R_0}{R}$$

$$\frac{dz}{dt} = - \frac{R_0}{R^2} \dot{R} = - (1+z) \frac{\dot{R}}{R} = - (1+z) H$$

$$\Omega_0 \equiv \Omega_{m,0} + \Omega_{r,0} + \Omega_{v,0}$$

$$H^2 = H_0^2 (\Omega_{m,0} (1+z)^3 + \Omega_{r,0} (1+z)^4 + \Omega_{v,0} + (1-\Omega_0) (1+z)^2)$$

$$\left(\frac{\dot{R}}{R}\right)^2 = \rightarrow$$

Since $z \gg 1$ for earlier times

$$\dot{R} \gg R$$

$\rightarrow$ so its expansion was faster at early times

$$dt = - \frac{dz}{(1+z) H(z)}$$

$$dt = \frac{-dz H_0^{-1}}{(1+z)} \left(\Omega_{m,0} (1+z)^3 + \Omega_{r,0} (1+z)^4 + \Omega_{v,0} + (1-\Omega_0) (1+z)^2 \right)^{-1/2}$$

* Current age of universe $z \rightarrow (0, \infty)$

$$t_{age} = - \int_{\infty}^0 () dz = \int_0^{\infty} \frac{dz}{(1+z) H(z)}$$

$$H_0 t_0 \approx \frac{2}{3} (0.7 \Omega_{m,0} - 0.3 \Omega_{v,0} + 0.3)^{-0.3}$$

Comoving Distance:

• photon: $(ds)^2 = 0 = (cdt)^2 - R^2 dr^2$

$$R dr = c dt$$

$$\Rightarrow R \frac{dr}{dz} = c \frac{dt}{dz} = c \left(\frac{-1}{(1+z) H(z)} \right)$$

• $R dr = \frac{-c dz}{(1+z) H(z)}$

$$\frac{R_0}{R} = (1+z) \Rightarrow R = \frac{R_0}{(1+z)}$$

$$\text{so, } \boxed{R_0 dr = - \frac{c dz}{H(z)}}$$

EdS: $\Omega_{m,0} = 1 = \Omega_0$, $\Omega_{v,0} = \Omega_{r,0} = 0$

$$\int R_0 dr = -\frac{cdz}{H} = \int_0^z \frac{-cdz}{H_0(1+z)^{3/2}} = \int_0^z$$

$$R_0 r_{ph}(z) = \left(\frac{2c}{H_0}\right) (1+z)^{-1/2} \Big|_0^z$$

• Missing = sign

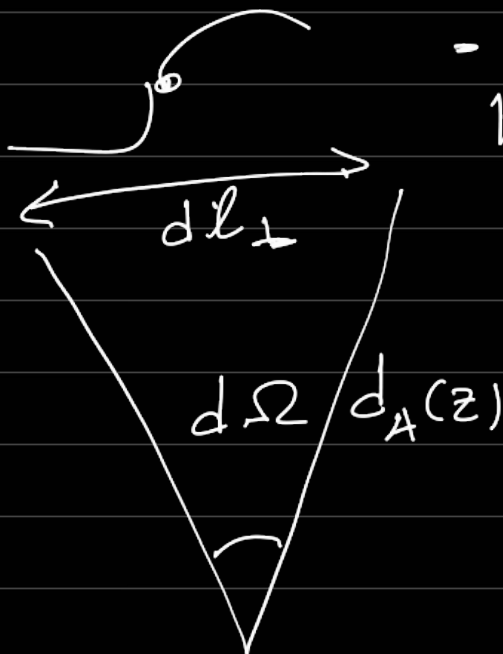
dS | $\Omega_0 = \Omega_{v,0} = 1$, $\Omega_{r,0} = \Omega_{m,0} = 0$

$$R_0 r(z) = \frac{cz}{H_0}$$

Angular Diameter Distance:

$$d\vec{s}^2 = c^2 dt^2 - R^2(t) (dr^2 + S_k^2(r) d\Omega^2)$$

$$d\Omega^2 = d\theta^2 + \sin^2(\theta) d\phi^2$$



$$= \underbrace{(Rdr)^2}_{\text{radial}} + \underbrace{(RS_k(r)d\Omega)^2}_{\text{angular}}$$

$$d\ell_{\perp} = R(t) S_k(r) d\Omega$$

• Proper transverse size: $d\ell_{\perp} = R(t) S_k(r) d\Omega$

$$d\ell_{\perp} = d\Omega d_A(z) = R(t) S_k(r) d\Omega$$

$$\frac{R_0 S_k(r)}{(1+z)}$$

$$\therefore d_A(z) = R(t) S_k(r) = \frac{R_0 S_k(r)}{(1+z)}$$

$$d_A(z) = \frac{R_0 S_k(r)}{(1+z)}$$

$$d_A \theta = \tilde{R}$$

* Note $S(r) = S(r(z))$

$$R_0 r(z) = \int_0^z \frac{c dz'}{H(z')}$$

i) EdS: $K=0$, $\Omega_{m,0} = 1$
 $H^2 = H_0^2 (1+z)^3$

$$H(z) = H_0 (1+z)^{3/2}$$

$$R_0 r(z) = \int_0^z \frac{c dz'}{H(z')}$$

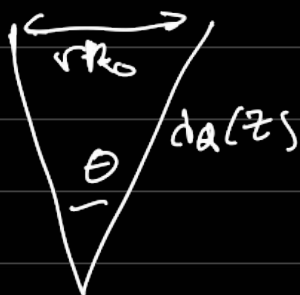
$$d_A(z) = \frac{R_0 r}{(1+z)} = \frac{zc}{H_0(1+z)} \left(1 - \frac{1}{\sqrt{1+z}} \right)$$

ii) dS:

$$d_A(z) = \frac{cz}{H_0(1+z)}$$

$$\rightarrow \frac{c}{H_0} \quad (z \rightarrow \infty)$$

B3. (c) $d_A(z) = \frac{zc}{H_0(1+z)} =$



$$d_A(z) \theta = dl$$

$$d_A(z) = \frac{dl}{\theta} = \frac{zc}{H_0(1+z)}$$

$$d_A \theta = dR_0 = \frac{zc}{H_0(1+z)} \theta = \frac{zc}{H_0(1+z)^{1/2}}$$

$$\theta =$$

Luminosity distance: $d_L(r) \Leftrightarrow S_r(r)$

$$I_r(r) dV = \frac{L_r(r) dV}{4\pi r^2}$$

• Normal eq were used to

(i) shift to source: γ 's come out over a surface area

$$4\pi (R S_K(r))^2$$

But we're at $z=0$ so $R=R_0$

$$S_{-A} = 4\pi R_0^2 S_K^2(r)$$

• time dilation: $dt \rightarrow (1+z) dt'$

• Redshifted Energy: $dE \rightarrow (1+z) dE$

• dV is reduced by $(1+z)$

• $N_0 \rightarrow (1+z) N_0$

$$S_r(N_0) = \frac{L_r((1+z)N_0)}{4\pi R_0^2 S_K^2(r) (1+z)}$$

$$S_{\text{tot}} = \int S_r(N_0) dN_0 = \frac{\int L_r(N) dN}{4\pi R_0^2 S_K^2(r) (1+z)^2}$$

$$S_{\text{tot}} = \frac{L_{\text{tot}}}{4\pi r^2} \quad \text{IF}$$

Luminosity distance: $d_L = R_0 S_K(r) (1+z)$

$$S_{\text{tot}} = \frac{L_{\text{tot}}}{4\pi d_L^2}$$

$$d_A(z) \doteq R_0 S_K(r) \frac{1}{1+z}$$

$$\frac{dL}{dA} = (1+z)^2 \Leftrightarrow dL = (1+z)^2 dA$$

Q2:

$$H^2 = H^2_0 - kc^2$$

$$H^2(1-\Omega) = \frac{1}{R^2} \quad H^2 = \frac{kc^2}{R^2} \frac{1}{\Omega-1}$$

$$R^2 = \frac{1}{H^2} \frac{kc^2}{(\Omega-1)}$$

$$R_0 = \frac{1}{H_0} \frac{kc^2}{(\Omega-1)}$$

Q3 $\Omega_0 = 1.1$

Q2:

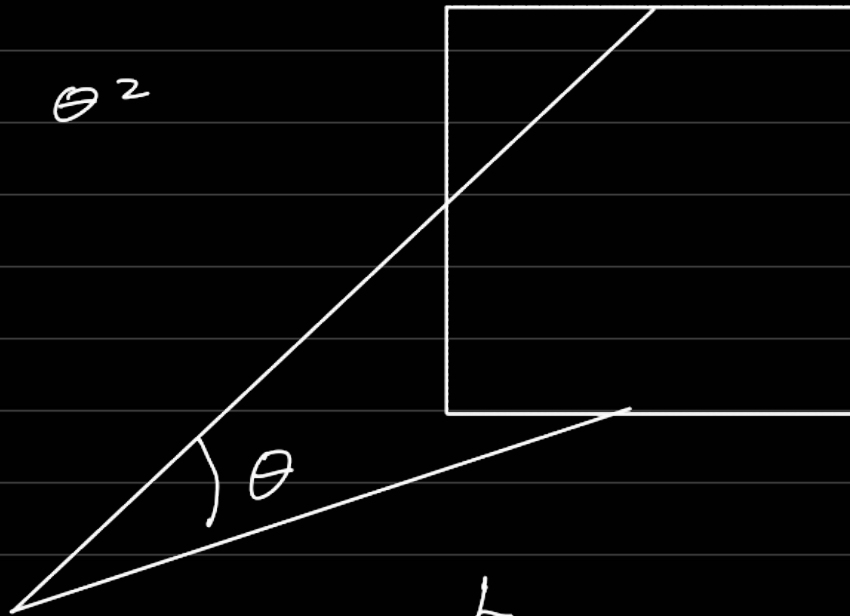


$$\theta r = d$$

$$S_{\text{tot}} = \frac{L_{\text{tot}}}{4\pi r^2}$$

• Solid angle: θ^2

$$r^2 = \frac{d^2}{\theta^2}$$



Flux: $S_r = \frac{F}{\theta^2} = \frac{\frac{L}{4\pi r^2}}{\theta^2} = \frac{L}{4\pi r^2 \theta^2}$

$$M_0 = \frac{L_{\text{tot}}}{4\pi d^2}$$

$$d_A = \frac{d}{\theta}$$

$$\theta = \frac{d}{d_A}$$

$$S = \frac{L_{\text{tot}}}{4\pi d_l^2} = \frac{L_{\text{tot}}}{4\pi R_0^2 S_n^2 (1+z)^2}$$

$$M(z) = \frac{S}{A^2} = \left(\frac{L_{\text{tot}}}{4\pi d_l^2} \right) \left(\frac{d_A}{d} \right)^2$$

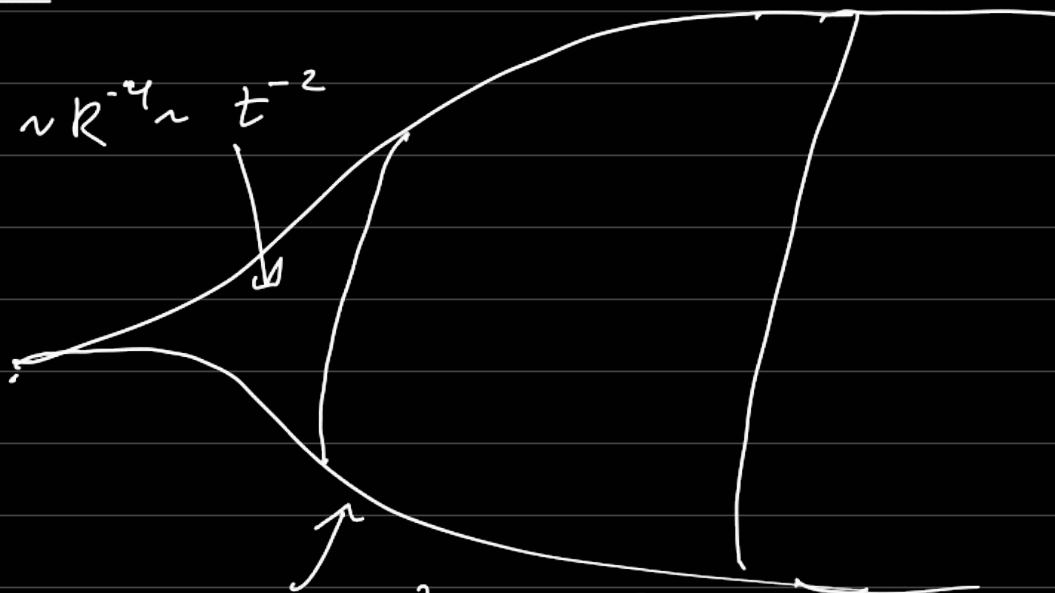
$$\mu(z) = \left(\frac{L_{tot}}{4\pi d^2} \right) \left(\frac{dA}{dL} \right)^2 = \frac{\mu_0}{(1+z)^4}$$

$d_L \equiv$ Lumin. Distance

d_A : Ang. Dist.

Thermal:

$$p_r \sim R^{-4} \sim t^{-2}$$



$$p_r + p_m \sim R^{-3} \sim t^{-2}$$

• However at early times:

$$p_r \sim n_\gamma R_\gamma \gg \text{Binding energies}$$

CMB: $p_r \rightarrow p_m$, black body

n_γ : thermal bath

$$\bullet \text{ Box } V = L^3 \quad / \quad dN = g(z\alpha)^3 d^3k$$

$$P(k) = \frac{e^{-E_n/k_B T}}{Z}$$

$$E_n^2 = \dots (n_x^2 + n_z^2 + n_y^2)$$

1

$$\ast \text{ Occupation \# : } f(\epsilon) = \frac{1}{(e^{-\mu/k_B T} \pm 1)}$$

1 p

1

∞ $\frac{1}{4\pi} \frac{d^2}{d^2}$

1

$$n = \frac{1}{V} \int f dN = g \frac{1}{(2\pi\hbar)^3} \int_0^\infty \frac{4\pi p^2}{e^{\epsilon(p)/kT} \pm 1} dp$$

• $p = \hbar k$

(i) $kT \gg mc^2$

$$E \approx pc$$

$$n = g \frac{1}{(2\pi\hbar)^3} \int_0^\infty \frac{4\pi p^2}{e^{pc/kT} \pm 1} dp$$

$$y \equiv \frac{pc}{kT} \Rightarrow dy = \frac{c}{kT} dp$$

• $dp = \frac{kT}{c} dy$ $p = \left(\frac{y kT}{c} \right)$

$$n_i = g \frac{1}{(2\pi\hbar)^3} \int_0^\infty 4\pi y^2 \left(\frac{kT}{c} \right)^2 \left(\frac{1}{e^y \pm 1} \right) \left(\frac{kT}{c} \right) dy$$

$$n \propto (kT)^3$$

(ii) $kT \ll mc^2$

$$E \sim \frac{p^2}{2m} + mc^2$$

$$E/kT \gg 1$$

$$n_{ii} = \frac{g}{(2\pi\hbar)^3} \int_0^\infty \frac{4\pi p^2}{\exp[(p^2/2m + mc^2)/kT]} dp$$

$$= \frac{g}{(2\pi\hbar)^3} \int_0^\infty e^{-mc^2/kT} e^{-p^2/2mkT} 4\pi p^2 dp$$

$$y = \frac{p}{\sqrt{2mkT}} \quad dp = \sqrt{2mkT} dy$$

$$n_{ii} = \frac{g}{(2\pi\hbar)^3} \int_0^\infty 4\pi 2mkT y^2 \sqrt{2mkT} e^{-y^2} e^{-mc^2/kT} dy$$

$$\propto g e^{-mc^2/kT} (kT)^{3/2}$$

